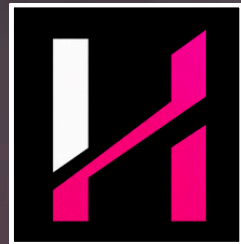


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INNOVA HACK CHAPTER-1

NATIONAL LEVEL HACKATHON

PROBLEM STATEMENT & GUIDELINES

ROUND 1 (ONLINE)

Timeline:

start:

25 July 2026
10:00 AM



Ends:

26 July 2026
10:00 AM

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HOW PROBLEM STATEMENTS WORK

Every domain offers two problem statements. Teams may choose either problem statement within their selected domain — pick the one that excites your team the most and plan to build a working, demoable solution around it. Your choice of problem statement does not need to be declared in advance; you may explore both during the online round and finalize your direction before submission.

DOMAIN 1: FINTECH

PROBLEM STATEMENT 1:

HIDDEN SUBSCRIPTION & RECURRING PAYMENT LEAK DETECTOR

BACKGROUND

Most people are quietly losing money every month to subscriptions they forgot about, price hikes they never noticed, and services they no longer use — buried inside cluttered bank statements, SMS alerts, and email inboxes that no one has time to review line by line.

OBJECTIVE

Build a system that scans a user's transaction history (bank statements, SMS alerts, or email notifications) to automatically detect every recurring subscription or payment. The system should flag silent price increases, identify subscriptions that have gone unused, and present a clear "leak score" along with a concrete action plan for each one - cancel, downgrade, or renegotiate.

SUGGESTED FOCUS AREAS

- Recurring-transaction detection from unstructured statement/SMS/email data
- Price-increase and anomaly detection over time
- "Leak score" with category-wise cost visualization
- Actionable, per-subscription cancellation or downgrade guidance

FINTECH

PROBLEM STATEMENT 2:

SMART EXPENSE & MICRO-INVESTMENT ASSISTANT

BACKGROUND:

Everyday spending data holds valuable insight, yet most people don't act on it. Round-up-and-invest models have shown that small, automated financial habits can compound into meaningful savings over time.

OBJECTIVE:

Build an application that tracks a user's spending, automatically categorizes transactions, and rounds up purchases to invest the spare change into a simulated portfolio (mutual funds, stocks, or crypto). Include a dashboard that visualizes savings growth over time along with simple, actionable spending insights.

SUGGESTED FOCUS AREAS:

- Automatic transaction categorization
- Round-up and simulated micro-investment engine
- Savings/growth visualization dashboard
- Personalized spending insights

DOMAIN 2: CYBERSECURITY

PROBLEM STATEMENT 1:

AUTONOMOUS THREAT HUNTER FOR INSIDER ATTACKS

BACKGROUND:

Insider threats are among the hardest to detect because the activity often resembles normal behavior right up until damage is done — an employee quietly accessing sensitive files before resignation, or data being moved out in small, disguised increments.

OBJECTIVE:

Build a system that ingests simulated organizational logs (login activity, file access, data transfers) and uses behavioral baselining to identify insider threats. The system should minimize false positives by relying on anomaly detection rather than static, rule-based triggers alone.

SUGGESTED FOCUS AREAS:

- Behavioral baseline modeling per user/entity
- Anomaly detection engine (not purely rule-based)
- Alert dashboard with severity scoring
- Low false-positive design philosophy

CYBERSECURITY

PROBLEM STATEMENT 2:

PHISHING & MALICIOUS URL DETECTOR

BACKGROUND:

Phishing remains one of the most common entry points for cyberattacks, and most users have no quick way to verify whether a link or email is safe before clicking.

OBJECTIVE

Build a browser extension or web application that analyzes URLs and emails in real time, flagging phishing attempts using a combination of heuristics (domain age, structural red flags, etc.) and a trained ML classifier. The tool should present a clear risk score along with a short explanation of why a link was flagged.

SUGGESTED FOCUS AREAS

- Real-time URL/email scanning
- Heuristic + ML-based classification
- Risk score with explanation
- Lightweight, user-friendly interface (extension or web app)

DOMAIN 3: GEN AI

PROBLEM STATEMENT 1:

AUTONOMOUS MULTI-AGENT RESEARCH & FACT-VERIFICATION SYSTEM

BACKGROUND

Generative AI tools are powerful researchers but often struggle with hallucination and unverified claims. A system where multiple AI agents check and challenge each other can produce far more trustworthy output than a single model working alone.

OBJECTIVE

Build a multi-agent pipeline where one agent researches a given topic, another cross-verifies claims against multiple sources, a third detects contradictions or hallucinations, and a final agent compiles a citation-backed report — complete with a confidence score for each claim.

SUGGESTED FOCUS AREAS

- Multi-agent orchestration (research, verification, synthesis)
- Contradiction / hallucination detection logic
- Source citation for every claim
- Per-claim confidence scoring.

GEN AI

PROBLEM STATEMENT 2:

PERSONALIZED AI STUDY / INTERVIEW COACH

BACKGROUND

Preparing for exams or interviews is often generic and one-size-fits-all, when what learners actually need is adaptive, personalized feedback based on their real strengths and gaps.

OBJECTIVE

Build a Gen AI-powered tool that takes a topic or job role as input, generates relevant practice questions, evaluates the user's spoken or written answers, and provides structured feedback — highlighting strengths, gaps, and suggested resources — while adapting difficulty based on performance.

SUGGESTED FOCUS AREAS

- Dynamic question generation by topic/role
- Answer evaluation (text and/or voice)
- Structured, actionable feedback
- Adaptive difficulty based on performance

DOMAIN 4: AGENTIC AI

PROBLEM STATEMENT 1:

AUTONOMOUS PERSONAL ASSISTANT AGENT FOR MULTI-STEP REAL-WORLD TASKS

BACKGROUND:

Everyday tasks — planning a trip, comparing options across services, coordinating a schedule — rarely take one step. They involve switching between apps, gathering information, making decisions, and following up, all of which people currently have to manage manually themselves.

OBJECTIVE:

Build an agent that can take a single high-level instruction (for example, "find and book the cheapest flight and hotel combo for next weekend under a set budget") and autonomously break it into sub-tasks, call multiple tools or APIs (real or simulated) to gather information, handle failures and retries along the way, ask for clarification only when truly necessary, and complete the task end-to-end with a clear summary of what it did and why.

SUGGESTED FOCUS AREAS:

- Task planning and decomposition
- Multi-tool/API orchestration with error handling
- Memory and state tracking across steps
- Transparent action log explaining the agent's decisions

AGENTIC AI

PROBLEM STATEMENT 2:

AI Meeting & Follow-Up Agent

BACKGROUND:

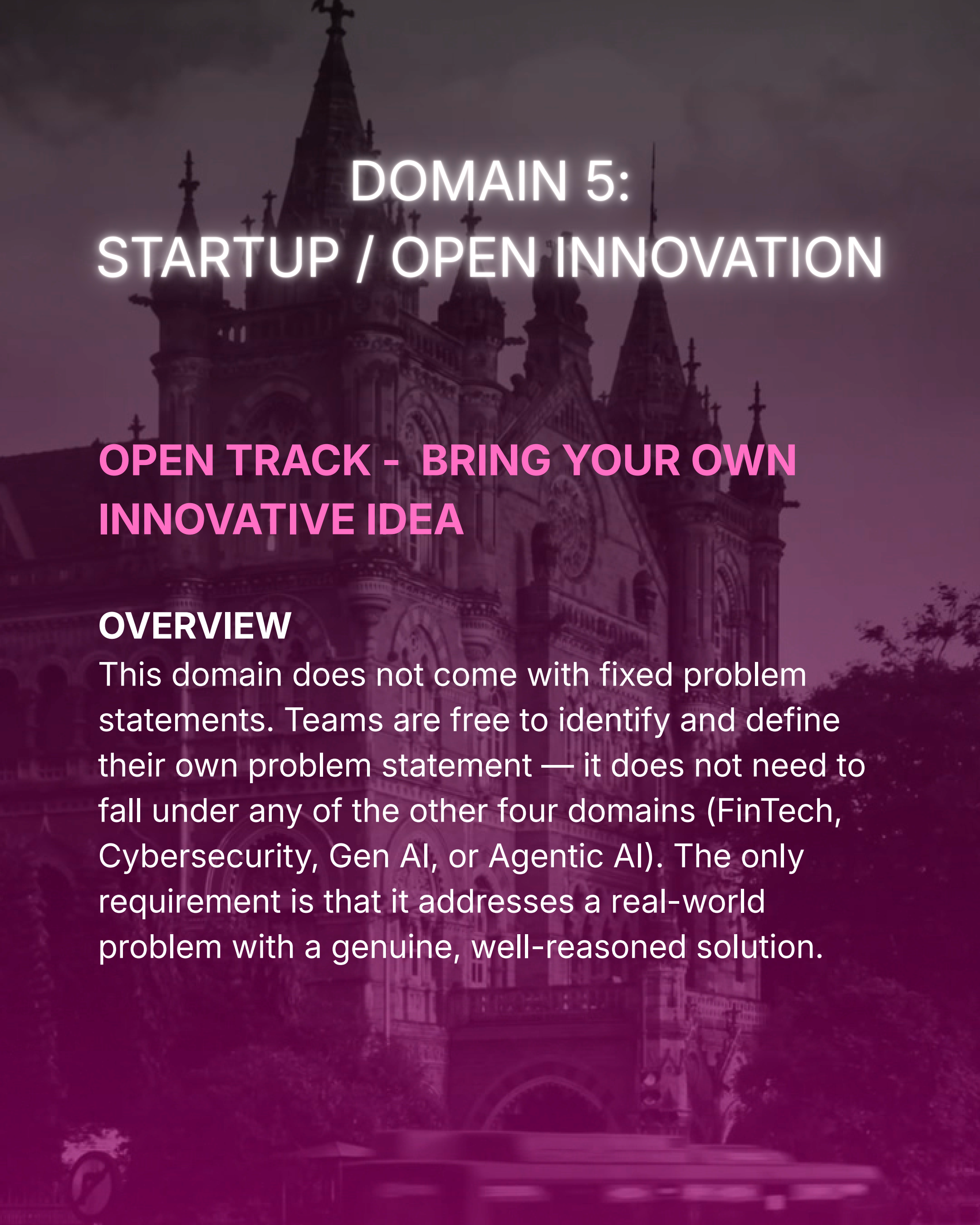
Meetings generate decisions and action items constantly, but the moment the call ends, most of it goes untracked — no one is quite sure who owes what, by when, and follow-ups rarely happen unless someone manually chases them.

OBJECTIVE:

Build an agent that processes a meeting transcript or recording, extracts key decisions and action items, assigns them to the right people, and autonomously sends reminders and follow-ups until each item is marked complete.

SUGGESTED FOCUS AREAS:

- Transcript/action-item extraction
- Owner and deadline assignment
- Automated reminder/follow-up loop
- Simple dashboard of pending vs. completed items



DOMAIN 5: STARTUP / OPEN INNOVATION

OPEN TRACK - BRING YOUR OWN INNOVATIVE IDEA

OVERVIEW

This domain does not come with fixed problem statements. Teams are free to identify and define their own problem statement — it does not need to fall under any of the other four domains (FinTech, Cybersecurity, Gen AI, or Agentic AI). The only requirement is that it addresses a real-world problem with a genuine, well-reasoned solution.

SUBMISSION GUIDELINES

SUBMISSION DEADLINE: 26 JULY 2026, 10:00 AM (IST)

- **Submit your project only through the official Google Form.**
- **Upload a deployed URL of your project**
- **Upload a (Google Drive link) containing:**
 2. Presentation (PPT) of 6–7 slides. (no template has been circulated)
 3. Optional: a 5-minute explanatory video demonstrating your project
- **Ensure the Drive link has public viewing access ("Anyone with the link can view").**
- **Clearly mention the following in your submission:**
 1. Team Name
 2. Team Leader's Name
 3. Team Members
 4. Selected Track / Problem Statement
- **Verify that all uploaded files are complete and accessible before submitting.**
- **Late submissions will not be accepted under any circumstances.**
- **Only one submission per team should be made by the Team Leader.**
- **incomplete, inaccessible, or incorrect submissions may be disqualified.**
- **Teams are advised to submit well before the deadline to avoid last-minute technical issues.**

PROJECT SUBMISSION FOR ROUND 1

(SCAN THE QR CODE OR USE THE LINK BELOW)



Submission link: <https://forms.gle/J41yUTNsrgbBUHhk37>